

JEE MAIN REVISION MASTER: PERIODIC TABLE & PERIODICITY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 173 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Ionization Enthalpy & Electron Gain Enthalpy	118 PYQs	Periodic Trends
Very High	Atomic & Ionic Radii Trends	67 PYQs	Radii

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Ionization Enthalpy & Electron Gain Enthalpy	$Z_{\text{eff}} = Z - \sigma, \quad \Delta_{\text{eg}}H(\text{Cl}) < \Delta_{\text{eg}}H(\text{F})$ (Cl is most negative in periodic table)
Atomic & Ionic Radii Trends	For isoelectronic: $r \propto \frac{1}{Z}$ implies $N^{3-} > O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ionization Enthalpy & Electron Gain Enthalpy (Very High Yield — 118 PYQs)

Core Principle: Periodic variations of ionization energy, exceptions due to half-filled/fully-filled subshells ($N > O$, $Be > B$), and electron gain enthalpy.

■ **Caution / JEE Pitfall:** Ionization enthalpy of Nitrogen is HIGHER than Oxygen due to stable half-filled $2p^3$ subshell ($N: 2s^2 2p^3 > O: 2s^2 2p^4$).

■ **Speed Trick:** Chlorine has MORE negative electron gain enthalpy than Fluorine because of compact size and high inter-electronic repulsion in $2p$ of Fluorine.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Match List - I with List - II. List - I Electronic configuration of neutral atom (where $n=2$) List - II 1st Ionization enthalpy." [Ans: A]
- [2026 Evening Shift] Q#1 (MCQ): "The 1st ionization enthalpy for Mg is +737 kJ/mol. The most probable estimated value of the 2nd ionization enthalpy..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "In period 4 of the periodic table, the elements with highest and lowest atomic radii are respectively...." [Ans: C]

3.2 Atomic & Ionic Radii Trends (Very High Yield — 67 PYQs)

Core Principle: Covalent, van der Waals, and ionic radii trends across periods and groups, and size shrinkage in isoelectronic ions.

■ **Caution / JEE Pitfall:** Noble gas radii are measured as van der Waals radii, making them appear abnormally larger than adjacent halogens.

■ **Speed Trick:** Due to lanthanoid contraction (poor shielding of 4f electrons), 4d and 5d transition elements have nearly identical radii (e.g. Zr $\approx$ Hf).

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "In period 4 of the periodic table, the elements with highest and lowest atomic radii are respectively...." **[Ans: C]**
- **[2026 Evening Shift] Q#3 (MCQ):** "Given below are two statements : Statement I : The number of pairs among $[Al_2O_3], [Cr_2O_3], [Cl_2O_7], [Mn_2O_7], [Na_2O], [V_2O_5]$..." **[Ans: A]**
- **[2026 Morning Shift] Q#3 (MCQ):** "Given below are two statements : Statement I : $K > Mg > Al > B$ is the correct order in terms of metallic character. Statement II : Atomic radius..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Match List - I with List - II. List - I Electronic configuration of neutral atom (where $n=2$) List - II 1st Ionization Energy (kJmol⁻¹) A. ns^2 I. 2080 B. ns^2np^1 II. 899 C. ns^2np^3 III. 800 D. ns^2np^6 IV. 1402 Choose the correct answer from the options given below :

- (A) -II, B-III, C-IV, D-I **[Correct]**
- (B) -IV, B-III, C-II, D-I
- (C) -III, B-II, C-IV, D-I
- (D) -III, B-II, C-I, D-IV

Correct Answer: A

Solution Analysis: Ionization energy is the energy needed to remove the most loosely held (outermost) electron from a neutral atom. Ionization energy generally increases when the effective nuclear charge increases. This happens because the nucleus attracts the valence electron more strongly, so more energy is required to remove it. Also, some electronic configurations are extra stable. Electrons in stable configurations are harder to remove, so their ionization energy is higher. Fully filled orbitals have greater ionisation energy due to stable electronic configuration. Order of I.E: $\text{fully-filled orbital}$

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Evening Shift (Q#3)

Given below are two statements : Statement I : The number of pairs among

$[\text{Al}_2\text{O}_3, \text{Cr}_2\text{O}_3], [\text{Cl}_2\text{O}_7, \text{Mn}_2\text{O}_7], [\text{Na}_2\text{O}, \text{V}_2\text{O}_5]$ and $[\text{CO}, \text{N}_2\text{O}]$ that contain oxides of same nature (acidic, basic, neutral or amphoteric) is 4 . Statement II : Among $\text{Na}_2\text{O}, \text{Al}_2\text{O}_3, \text{CO}$ and Cl_2O_7 , the most basic and acidic oxides are Na_2O and Cl_2O_7 , respectively. In the light of the above statements, choose the correct answer from the options given below :

- (A) Both Statement I and Statement II are true **[Correct]**
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Correct Answer: A

Solution Analysis: Evaluating Statement I: We need to determine the chemical nature of each pair of oxides: Pair 1:

$[\text{Al}_2\text{O}_3, \text{Cr}_2\text{O}_3]$ Al_2O_3 (Aluminium oxide) is a standard example of an amphoteric oxide as it reacts with both acids and bases. Cr_2O_3 (Chromium(III) oxide) has chromium in the +3 oxidation state. Transition metal oxides in intermediate oxidation states are generally amphoteric. Both oxides are amphoteric, so they have the same nature. Pair 2: $[\text{Cl}_2\text{O}_7, \text{Mn}_2\text{O}_7]$ Cl_2O_7 is a non-metal oxide in a very high oxidation state (+7), which makes it strongly acidic.